

“PAPER_{0:D3}” -f [int]# PAPER #93 □ M87* Event Horizon: UQFF Field Analysis

Title: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ([SCm] □ 0.99, [SSq] = 0.57)

Date: March 7, 2026

Source Data: validate_uqff_muge.py, from_system('M87'), EHT 2019-2024 data

Index Slot: §1.12 UQFF Master Calculators,

\$n = [int]# PAPER #93 □ M87* Event Horizon: UQFF Field Analysis

Title: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ([SCm] □ 0.99, [SSq] = 0.57)

Date: March 7, 2026

Source Data: validate_uqff_muge.py, from_system('M87'), EHT 2019-2024 data

Index Slot: §1.12 UQFF Master Calculators, PAPER_093

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_\odot$, $d = 16.8$ Mpc), provides a strong-field test of UQFF at a mass $1,600\times$ Sgr A. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{\text{UQFF}} = 0.99 T_{\text{H}} = 1.34 \times 10^{17}$ K. All 8 terms are finite and g_{total} is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($\alpha = 5.0 \times 10^{-4}$ day $^{-1}$, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. M87* System Parameters

Parameter	Value	Source
M_{BH}	$1.26 \times 10^6 \text{ kg}$ ($6.5 \times 10^6 M_\odot$)	EHT 2019 (first image)
$r_{\text{Schwarzschild}}$	$1.92 \times 10^{13} \text{ m}$	$2GM/c^2$
$r_{\text{horizon (UQFF)}}$	$1.95 \times 10^{13} \text{ m}$	$r_{\text{S}} \times (1 + 0.015)$
Distance	$16.8 \text{ Mpc} = 5.18 \times 10^{23} \text{ m}$	Virgo Cluster
Spin (a/M)	0.90 ± 0.05	EHT 2024
Jet power P_{jet}	$\sim 1044 \text{ erg/s}$	VLA/VLBI
T_{H} (GR)	$1.35 \times 10^{17} \text{ K}$	$\alpha c^3/(8\pi GM_{\text{BH}})$

Parameter	Value	Source
T_UQFF	$1.34 \times 10^{17} \text{ K}$	$T_H \times 0.99$

2. 8-Term MUGE at $r_{\text{horizon}} = 1.95 \times 10^{13} \text{ m}$

Term	Value (m/s ²)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	Ug4 ? $M^2/r6$? M87 larger r offsets large M^2
U_i	0.14	
cosmological	-9.1×10^{21}	? negligible at horizon
quantum	$+2.0 \times 10^{41}$	Planck-scale
fluid	$+6.2 \times 10^{13}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, » far_field
g_total	2211	100%

Newtonian $g_{\text{Newton}} = 2207 \text{ m/s}^2$. MUGE total = 2211 m/s² ? UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $G \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With $\eta = \eta_{\text{acc}}/\eta_{\text{Edd}} \approx 10^{-3}$ (low state) and $[\text{SCm}] = 0.99$:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $\eta_{\text{jet}} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $\theta_{\text{ring}} = 42 \pm 3 \mu\text{as}$? physical $r_{\text{ring}} = 5.0 \text{ GM}/c^2$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via $[\text{SCm}]$:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [\text{SCm}]}{2}} = r_{\text{GR}} \times \sqrt{1.005} \approx 1.0025 r_{\text{GR}}$$

?? = +0.25% ? 0.1 μas shift (undetectable by current EHT at ±3 μas precision).

5. Coherence vs Distance

At M87* with its much larger r_horizon (vs Sgr A*):

Location	r/r_horizon	g_coh	Ratio
At horizon (1.95×10 ¹³ m)	1.0	g_coh,0	1.000
1 kpc (3.1×10 ¹⁷ m)	1.6×10 ⁶	~0	~10 ⁻⁶

From validator: assert coh_at_horizon > coh_far * 1e6 □ **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_H^{\text{M87*}} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\text{UQFF}}^{\text{M87*}} = 0.9999 \times T_H = 1.34 \times 10^{-17} \text{ K}$$

? = 0.01% reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	? PASS
g_total = Newton + 0.18%	? PASS
No NaN/Inf for M87*	? PASS
Coherence peak at horizon	? PASS
Jet power UQFF estimate	3.6×10 ⁴⁴ erg/s (consistent)
Shadow diameter deviation	0.25% (« EHT precision)
T_UQFF	1.34 × 10 ⁻¹⁷ K

Source: validate_uqff_muge.py | from system('M87') | EHT 2019-2024 | [SCm]=0.99
| all 8 terms PASS .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_\odot$, $d = 16.8$ Mpc), provides a strong-field test of UQFF at a mass $1,600\times$ Sgr A. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{\text{UQFF}} = 0.99 T_{\text{H}} = 1.34 \times 10^{17}$ K. All 8 terms are finite and g_{total} is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($\alpha = 5.0 \times 10^{-4}$ day $^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. M87* System Parameters

Parameter	Value	Source
M_{BH}	$1.26 \times 10^6 \text{ kg}$ ($6.5 \times 10^6 M_\odot$)	EHT 2019 (first image)
$r_{\text{Schwarzschild}}$	$1.92 \times 10^{13} \text{ m}$	$2GM/c^2$
r_{horizon} (UQFF)	$1.95 \times 10^{13} \text{ m}$	$r_{\text{S}} \times (1 + 0.015)$
Distance	$16.8 \text{ Mpc} = 5.18 \times 10^{23} \text{ m}$	Virgo Cluster
Spin (a/M)	0.90 ± 0.05	EHT 2024
Jet power P_{jet}	$\sim 1044 \text{ erg/s}$	VLA/VLBI
T_{H} (GR)	$1.35 \times 10^{17} \text{ K}$	$\hbar c^3/(8\pi G M_{\text{BH}})$
T_{UQFF}	$1.34 \times 10^{17} \text{ K}$	$T_{\text{H}} \times 0.99$

2. 8-Term MUGE at $r_{\text{horizon}} = 1.95 \times 10^{13} \text{ m}$

Term	Value (m/s 2)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	$Ug_4 \propto M^2/r_6 \propto M_{\text{BH}}$ larger r offsets large M^2
U_i	0.14	
cosmological	-9.1×10^{-21}	$\propto$ negligible at horizon
quantum	$+2.0 \times 10^{-41}$	Planck-scale
fluid	$+6.2 \times 10^{-13}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, $\gg$ far_field
g_{total}	2211	100%

Newtonian $g_{\text{Newton}} = 2207 \text{ m/s}^2$. MUGE total = $2211 \text{ m/s}^2 \propto$ UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $\Gamma \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With $\eta = \eta_{\text{acc}}/\eta_{\text{Edd}} \approx 10^{-3}$ (low state) and $[\text{SCm}] = 0.99$:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $\eta_{\text{jet}} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $\theta_{\text{ring}} = 42 \pm 3 \mu\text{as}$ $\Rightarrow$ physical $r_{\text{ring}} = 5.0 \text{ GM}/c^2$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via $[\text{SCm}]$:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [\text{SCm}]}{2}} = r_{\text{GR}} \times \sqrt{1.005} \approx 1.0025 r_{\text{GR}}$$

$\Delta\theta = +0.25\% \Rightarrow 0.1 \mu\text{as}$ shift (undetectable by current EHT at $\pm 3 \mu\text{as}$ precision).

5. Coherence vs Distance

At M87* with its much larger r_{horizon} (vs Sgr A*):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon ($1.95 \times 10^{13} \text{ m}$)	1.0	$g_{\text{coh},0}$	1.000
1 kpc ($3.1 \times 10^{17} \text{ m}$)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_H^{M87*} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\text{UQFF}}^{M87*} = 0.9999 \times T_H = 1.34 \times 10^{-17} \text{ K}$$

? = 0.01% reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	? PASS
g_total = Newton + 0.18%	? PASS
No NaN/Inf for M87*	? PASS
Coherence peak at horizon	? PASS
Jet power UQFF estimate	3.6×10 ⁴⁴ erg/s (consistent)
Shadow diameter deviation	0.25% (« EHT precision)
T_UQFF	1.34 × 10 ⁻¹⁷ K

Source: `validate_uqff_muge.py` | `from_system('M87')` | EHT 2019-2024 | [SCm]=0.99
| all 8 terms PASS .Groups[1].Value □ M87* Event Horizon: UQFF Field Analysis

Title: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ([SCm] □ 0.99, [SSq] = 0.57)

Date: March 7, 2026

Source Data: `validate_uqff_muge.py`, `from_system('M87')`, EHT 2019-2024 data

Index Slot: §1.12 UQFF Master Calculators,

\$n = [int]# "PAPER_{0:D3}" -f [int]# PAPER #93 □ M87* Event Horizon: UQFF Field Analysis

Title: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ([SCm] □ 0.99, [SSq] = 0.57)

Date: March 7, 2026

Source Data: `validate_uqff_muge.py`, `from_system('M87')`, EHT 2019-2024 data

Index Slot: §1.12 UQFF Master Calculators,

\$n = [int]# PAPER #93 □ M87* Event Horizon: UQFF Field Analysis

Title: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ([SCm] = 0.99, [SSq] = 0.57)

Date: March 7, 2026

Source Data: validate_uqff_muge.py, from_system('M87'), EHT 2019-2024 data

Index Slot: §1.12 UQFF Master Calculators, PAPER_093

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_\odot$, $d = 16.8$ Mpc), provides a strong-field test of UQFF at a mass $1,600 \times$ Sgr A. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{\text{UQFF}} = 0.99 T_{\text{H}} = 1.34 \times 10^{17}$ K. All 8 terms are finite and g_{total} is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($\beta = 5.0 \times 10^{-4}$ day $^{-1}$, [SSq] = 0.57) uniquely enabling this analysis by establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. M87* System Parameters

Parameter	Value	Source
M_BH	$1.26 \times 10^6 \text{ kg}$ ($6.5 \times 10^6 M_\odot$)	EHT 2019 (first image)
r_Schwarzschild	$1.92 \times 10^{13} \text{ m}$	$2GM/c^2$
r_horizon (UQFF)	$1.95 \times 10^{13} \text{ m}$	$r_S \times (1 + 0.015)$
Distance	$16.8 \text{ Mpc} = 5.18 \times 10^{23} \text{ m}$	Virgo Cluster
Spin (a/M)	0.90 ± 0.05	EHT 2024
Jet power P_jet	$\sim 10^{44} \text{ erg/s}$	VLA/VLBI
T_H (GR)	$1.35 \times 10^{17} \text{ K}$	$\hbar c^3/(8\pi G M k_B)$
T_UQFF	$1.34 \times 10^{17} \text{ K}$	$T_H \times 0.99$

2. 8-Term MUGE at r_horizon = $1.95 \times 10^{13} \text{ m}$

Term	Value (m/s ²)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	$Ug_4 \approx M^2/r_6 \approx$ M87 larger r offsets large M^2
U_i	0.14	
cosmological	-9.1×10^{-21}	$\approx$ negligible at horizon

Term	Value (m/s ²)	Notes
quantum	$+2.0 \times 10^{41}$	Planck-scale
fluid	$+6.2 \times 10^{13}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, » far_field
g_total	2211	100%

Newtonian $g_{\text{Newton}} = 2207 \text{ m/s}^2$. MUGE total = 2211 m/s^2 ? UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $G \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With $\eta = \eta_{\text{acc}}/\eta_{\text{Edd}} \approx 10^{-3}$ (low state) and $[\text{SCm}] = 0.99$:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $\eta_{\text{jet}} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $\theta_{\text{ring}} = 42 \pm 3 \mu\text{as}$? physical $r_{\text{ring}} = 5.0 \text{ GM}/c^2$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via $[\text{SCm}]$:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [\text{SCm}]}{2}} = r_{\text{GR}} \times \sqrt{1.005} \approx 1.0025 r_{\text{GR}}$$

$\Delta\theta = +0.25\% \approx 0.1 \mu\text{as}$ shift (undetectable by current EHT at $\pm 3 \mu\text{as}$ precision).

5. Coherence vs Distance

At M87* with its much larger r_{horizon} (vs Sgr A*):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon (1.95×10^{13} m)	1.0	$g_{\text{coh},0}$	1.000
1 kpc (3.1×10^{17} m)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_H^{\text{M87*}} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\text{UQFF}}^{\text{M87*}} = 0.9999 \times T_H = 1.34 \times 10^{-17} \text{ K}$$

$\delta = 0.01\%$ reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	δ PASS
$g_{\text{total}} = \text{Newton} + 0.18\%$	δ PASS
No NaN/Inf for M87*	δ PASS
Coherence peak at horizon	δ PASS
Jet power UQFF estimate	3.6×10^{44} erg/s (consistent)
Shadow diameter deviation	0.25% ($\ll$ EHT precision)
T_{UQFF}	1.34×10^{-17} K

Source: `validate_uqff_muge.py` | `from_system('M87')` | EHT 2019-2024 | $[SCM]=0.99$
 | all 8 terms PASS .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_\odot$, $d = 16.8$ Mpc), provides a strong-field test of UQFF at a mass $1,600\times$ Sgr A. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{\text{UQFF}} = 0.99 T_{\text{H}} = 1.34 \times 10^{17}$ K. All 8 terms are finite and g_{total} is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day $^{1/2}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. M87* System Parameters

Parameter	Value	Source
M_BH	$1.26 \times 10^6 \text{ kg}$ ($6.5 \times 10^6 M_\odot$)	EHT 2019 (first image)
r_Schwarzschild	$1.92 \times 10^{13} \text{ m}$	$2GM/c^2$
r_horizon (UQFF)	$1.95 \times 10^{13} \text{ m}$	$r_S \times (1 + 0.015)$
Distance	$16.8 \text{ Mpc} = 5.18 \times 10^{23} \text{ m}$	Virgo Cluster
Spin (a/M)	0.90 ± 0.05	EHT 2024
Jet power P_jet	$\sim 1044 \text{ erg/s}$	VLA/VLBI
T_H (GR)	$1.35 \times 10^{17} \text{ K}$	$\tau c^3/(8\pi GMk_B)$
T_UQFF	$1.34 \times 10^{17} \text{ K}$	$T_H \times 0.99$

2. 8-Term MUGE at r_horizon = $1.95 \times 10^{13} \text{ m}$

Term	Value (m/s ²)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	$Ug_4 \approx M^2/r_6 \approx M87$ larger r offsets large M^2
U_i	0.14	
cosmological	-9.1×10^{21}	$\approx$ negligible at horizon
quantum	$+2.0 \times 10^{24}$	Planck-scale
fluid	$+6.2 \times 10^{13}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, $\gg$ far_field
g_total	2211	100%

Newtonian $g_{\text{Newton}} = 2207 \text{ m/s}^2$. MUGE total = $2211 \text{ m/s}^2 \approx$ UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $\Gamma \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With $\eta = \eta_{\text{acc}}/\eta_{\text{Edd}} \approx 10^{-3}$ (low state) and $[\text{SCm}] = 0.99$:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $\eta_{\text{jet}} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $\theta_{\text{ring}} = 42 \pm 3 \mu\text{as}$ $\Rightarrow$ physical $r_{\text{ring}} = 5.0 \text{ GM}/c^2$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via $[\text{SCm}]$:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [\text{SCm}]}{2}} = r_{\text{GR}} \times \sqrt{1.005} \approx 1.0025 r_{\text{GR}}$$

$\Delta\theta = +0.25\% \approx 0.1 \mu\text{as}$ shift (undetectable by current EHT at $\pm 3 \mu\text{as}$ precision).

5. Coherence vs Distance

At M87* with its much larger r_{horizon} (vs Sgr A*):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon ($1.95 \times 10^{13} \text{ m}$)	1.0	$g_{\text{coh},0}$	1.000
1 kpc ($3.1 \times 10^{17} \text{ m}$)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_H^{M87*} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\text{UQFF}}^{M87*} = 0.9999 \times T_H = 1.34 \times 10^{-17} \text{ K}$$

? = 0.01% reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	? PASS
g_total = Newton + 0.18%	? PASS
No NaN/Inf for M87*	? PASS
Coherence peak at horizon	? PASS
Jet power UQFF estimate	3.6×10 ⁴⁴ erg/s (consistent)
Shadow diameter deviation	0.25% (« EHT precision)
T_UQFF	1.34 × 10 ⁻¹⁷ K

Source: `validate_uqff_muge.py` | `from_system('M87')` | EHT 2019-2024 | [SCm]=0.99
| all 8 terms PASS .Groups[1].Value □ M87* Event Horizon: UQFF Field Analysis

Title: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ([SCm] □ 0.99, [SSq] = 0.57)

Date: March 7, 2026

Source Data: `validate_uqff_muge.py`, `from_system('M87')`, EHT 2019-2024 data

Index Slot: §1.12 UQFF Master Calculators, "PAPER_{0:D3}" -f [int]# PAPER #93
□ M87* Event Horizon: UQFF Field Analysis

Title: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ([SCm] □ 0.99, [SSq] = 0.57)

Date: March 7, 2026

Source Data: `validate_uqff_muge.py`, `from_system('M87')`, EHT 2019-2024 data

Index Slot: §1.12 UQFF Master Calculators,
\$n = [int]# PAPER #93 □ M87* Event Horizon: UQFF Field Analysis

Title: M87* Event Horizon UQFF Field Analysis: 8-Term MUGE Gravity and Relativistic Jet Coupling

Author: Daniel T. Murphy

Framework: UQFF Star-Magic ([SCm] = 0.99, [SSq] = 0.57)

Date: March 7, 2026

Source Data: validate_uqff_muge.py, from_system('M87'), EHT 2019-2024 data

Index Slot: §1.12 UQFF Master Calculators, PAPER_093

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_\odot$, $d = 16.8$ Mpc), provides a strong-field test of UQFF at a mass $1,600 \times$ Sgr A. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{\text{UQFF}} = 0.99 T_{\text{H}} = 1.34 \times 10^{17}$ K. All 8 terms are finite and g_{total} is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($\alpha = 5.0 \times 10^{-4}$ day $^{-1}$, [SSq] = 0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. M87* System Parameters

Parameter	Value	Source
M_BH	$1.26 \times 10^6 \text{ kg}$ ($6.5 \times 10^6 M_\odot$)	EHT 2019 (first image)
r_Schwarzschild	$1.92 \times 10^{13} \text{ m}$	$2GM/c^2$
r_horizon (UQFF)	$1.95 \times 10^{13} \text{ m}$	$r_S \times (1 + 0.015)$
Distance	$16.8 \text{ Mpc} = 5.18 \times 10^{23} \text{ m}$	Virgo Cluster
Spin (a/M)	0.90 ± 0.05	EHT 2024
Jet power P_jet	$\sim 10^{44} \text{ erg/s}$	VLA/VLBI
T_H (GR)	$1.35 \times 10^{17} \text{ K}$	$\hbar c^3/(8\pi G M k_B)$
T_UQFF	$1.34 \times 10^{17} \text{ K}$	$T_H \times 0.99$

2. 8-Term MUGE at r_horizon = $1.95 \times 10^{13} \text{ m}$

Term	Value (m/s ²)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	$Ug_4 \approx M^2/r_6 \approx$ M87 larger r offsets large M^2
U_i	0.14	
cosmological	-9.1×10^{-21}	$\approx$ negligible at horizon

Term	Value (m/s ²)	Notes
quantum	$+2.0 \times 10^{41}$	Planck-scale
fluid	$+6.2 \times 10^{13}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, » far_field
g_total	2211	100%

Newtonian $g_{\text{Newton}} = 2207 \text{ m/s}^2$. MUGE total = 2211 m/s² ? UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $G \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With $\eta = \eta_{\text{acc}}/\eta_{\text{Edd}} \approx 10^{-3}$ (low state) and $[\text{SCm}] = 0.99$:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $\eta_{\text{jet}} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $\theta_{\text{ring}} = 42 \pm 3 \mu\text{as}$? physical $r_{\text{ring}} = 5.0 \text{ GM}/c^2$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via $[\text{SCm}]$:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [\text{SCm}]}{2}} = r_{\text{GR}} \times \sqrt{1.005} \approx 1.0025 r_{\text{GR}}$$

$\Delta\theta = +0.25\% \approx 0.1 \mu\text{as}$ shift (undetectable by current EHT at $\pm 3 \mu\text{as}$ precision).

5. Coherence vs Distance

At M87* with its much larger r_{horizon} (vs Sgr A*):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon (1.95×10^{13} m)	1.0	$g_{\text{coh},0}$	1.000
1 kpc (3.1×10^{17} m)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_H^{\text{M87*}} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\text{UQFF}}^{\text{M87*}} = 0.9999 \times T_H = 1.34 \times 10^{-17} \text{ K}$$

$\delta = 0.01\%$ reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	δ PASS
$g_{\text{total}} = \text{Newton} + 0.18\%$	δ PASS
No NaN/Inf for M87*	δ PASS
Coherence peak at horizon	δ PASS
Jet power UQFF estimate	3.6×10^{44} erg/s (consistent)
Shadow diameter deviation	0.25% ($\ll$ EHT precision)
T_{UQFF}	1.34×10^{-17} K

Source: `validate_uqff_muge.py` | `from_system('M87')` | EHT 2019-2024 | $[SCM]=0.99$
 | all 8 terms PASS .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_\odot$, $d = 16.8$ Mpc), provides a strong-field test of UQFF at a mass $1,600\times$ Sgr A. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{\text{UQFF}} = 0.99 T_{\text{H}} = 1.34 \times 10^{17}$ K. All 8 terms are finite and g_{total} is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($\tau = 5.0 \times 10^4$ day $^{1/2}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. M87* System Parameters

Parameter	Value	Source
M_BH	$1.26 \times 10^6 \text{ kg}$ ($6.5 \times 10^6 M_\odot$)	EHT 2019 (first image)
r_Schwarzschild	$1.92 \times 10^{13} \text{ m}$	$2GM/c^2$
r_horizon (UQFF)	$1.95 \times 10^{13} \text{ m}$	$r_S \times (1 + 0.015)$
Distance	$16.8 \text{ Mpc} = 5.18 \times 10^{23} \text{ m}$	Virgo Cluster
Spin (a/M)	0.90 ± 0.05	EHT 2024
Jet power P_jet	$\sim 1044 \text{ erg/s}$	VLA/VLBI
T_H (GR)	$1.35 \times 10^{17} \text{ K}$	$\tau c^3/(8\pi GMk_B)$
T_UQFF	$1.34 \times 10^{17} \text{ K}$	$T_H \times 0.99$

2. 8-Term MUGE at r_horizon = $1.95 \times 10^{13} \text{ m}$

Term	Value (m/s ²)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	$Ug4 \propto M^2/r^6 \propto M87$ larger r offsets large M^2
U_i	0.14	
cosmological	-9.1×10^{21}	$\propto$ negligible at horizon
quantum	$+2.0 \times 10^{14}$	Planck-scale
fluid	$+6.2 \times 10^{13}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, $\gg$ far_field
g_total	2211	100%

Newtonian $g_{\text{Newton}} = 2207 \text{ m/s}^2$. MUGE total = $2211 \text{ m/s}^2 \propto$ UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $\Gamma \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With $\eta = \eta_{\text{acc}}/\eta_{\text{Edd}} \approx 10^{-3}$ (low state) and $[\text{SCm}] = 0.99$:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $\eta_{\text{jet}} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $\theta_{\text{ring}} = 42 \pm 3 \mu\text{as}$ $\Rightarrow$ physical $r_{\text{ring}} = 5.0 \text{ GM}/c^2$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via $[\text{SCm}]$:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [\text{SCm}]}{2}} = r_{\text{GR}} \times \sqrt{1.005} \approx 1.0025 r_{\text{GR}}$$

$\Delta\theta = +0.25\% \approx 0.1 \mu\text{as}$ shift (undetectable by current EHT at $\pm 3 \mu\text{as}$ precision).

5. Coherence vs Distance

At M87* with its much larger r_{horizon} (vs Sgr A*):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon ($1.95 \times 10^{13} \text{ m}$)	1.0	$g_{\text{coh},0}$	1.000
1 kpc ($3.1 \times 10^{17} \text{ m}$)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_H^{M87*} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\text{UQFF}}^{M87*} = 0.9999 \times T_H = 1.34 \times 10^{-17} \text{ K}$$

? = 0.01% reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	? PASS
g_total = Newton + 0.18%	? PASS
No NaN/Inf for M87*	? PASS
Coherence peak at horizon	? PASS
Jet power UQFF estimate	3.6×10 ⁴⁴ erg/s (consistent)
Shadow diameter deviation	0.25% (« EHT precision)
T_UQFF	1.34 × 10 ⁻¹⁷ K

Source: `validate_uqff_muge.py` | `from_system('M87')` | EHT 2019-2024 | [SCm]=0.99
| all 8 terms PASS .Groups[1].Value

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_\odot$, $d = 16.8 \text{ Mpc}$), provides a strong-field test of UQFF at a mass $1,600 \times \text{Sgr A}$. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{\text{UQFF}} = 0.99 T_H = 1.34 \times 10^{-17} \text{ K}$. All 8 terms are finite and `g_total` is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^4 \text{ day}^{-1}$, [SSq] = 0.57) uniquely enabling this analysis □ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. M87* System Parameters

Parameter	Value	Source
M_BH	$1.26 \times 10^4 \text{ kg}$ ($6.5 \times 10^? \text{ M?}$)	EHT 2019 (first image)
r_Schwarzschild	$1.92 \times 10^{13} \text{ m}$	$2GM/c^2$
r_horizon (UQFF)	$1.95 \times 10^{13} \text{ m}$	$r_S \times (1 + 0.015)$
Distance	$16.8 \text{ Mpc} = 5.18 \times 10^{23} \text{ m}$	Virgo Cluster
Spin (a/M)	0.90 ± 0.05	EHT 2024
Jet power P_jet	$\sim 1044 \text{ erg/s}$	VLA/VLBI
T_H (GR)	$1.35 \times 10^{?17} \text{ K}$	$?c^3/(8pGMk_B)$
T_UQFF	$1.34 \times 10^{?17} \text{ K}$	$T_H \times 0.99$

2. 8-Term MUGE at r_horizon = $1.95 \times 10^{13} \text{ m}$

Term	Value (m/s ²)	Notes
base_gravity	2207	Newton dominant
sum_Ug	3.75	Ug4 ? M ² /r6 ? M87 larger r offsets large M ²
U_i	0.14	
cosmological	$-9.1 \times 10^{?21}$	? negligible at horizon
quantum	$+2.0 \times 10^{?41}$	Planck-scale
fluid	$+6.2 \times 10^{?13}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, » far_field
g_total	2211	100%

Newtonian g_Newton = 2207 m/s². MUGE total = 2211 m/s² ? UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $G \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With $? = ?_{\text{acc}}/?_{\text{Edd}} \approx 10^{-3}$ (low state) and $[\text{SCm}] = 0.99$:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $?_{\text{jet}} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $r_{\text{ring}} = 42 \pm 3 \mu\text{as}$? physical $r_{\text{ring}} = 5.0 \text{ GM}/c^2$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via [SCm]:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [\text{SCm}]}{2}} = r_{\text{GR}} \times \sqrt{1.005} \approx 1.0025 r_{\text{GR}}$$

?? = +0.25% ? 0.1 μas shift (undetectable by current EHT at $\pm 3 \mu\text{as}$ precision).

5. Coherence vs Distance

At M87* with its much larger r_{horizon} (vs Sgr A*):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon ($1.95 \times 10^{13} \text{ m}$)	1.0	$g_{\text{coh},0}$	1.000
1 kpc ($3.1 \times 10^{17} \text{ m}$)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_H^{\text{M87*}} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\text{UQFF}}^{\text{M87*}} = 0.9999 \times T_H = 1.34 \times 10^{-17} \text{ K}$$

? = 0.01% reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	? PASS
$g_{\text{total}} = \text{Newton} + 0.18\%$	? PASS

Validation	Result
No NaN/Inf for M87*	? PASS
Coherence peak at horizon	? PASS
Jet power UQFF estimate	3.6×10^{44} erg/s (consistent)
Shadow diameter deviation	0.25% ($\ll$ EHT precision)
T_UQFF	1.34×10^{17} K

Source: `validate_uqff_muge.py | from_system('M87') | EHT 2019-2024 | [SCm]=0.99`
| all 8 terms PASS .Groups[1].Value "PAPER_{0:D3}" -f \$n

Abstract

M87, the first black hole imaged by the Event Horizon Telescope ($M = 6.5 \times 10^6 M_\odot$, $d = 16.8$ Mpc), provides a strong-field test of UQFF at a mass $1,600 \times$ Sgr A. The `from_system('M87')` constructor in `validate_uqff_muge.py` encodes EHT parameters and computes the 8-term MUGE field, jet power (Ug3-mediated), and Hawking temperature $T_{\text{UQFF}} = 0.99 T_{\text{H}} = 1.34 \times 10^{17}$ K. All 8 terms are finite and g_{total} is consistent with the VLBI ring diameter.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^{-4}$ day $^{-1}$, $[\text{SSq}] = 0.57$) uniquely enabling this analysis $\square$ establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. M87* System Parameters

Parameter	Value	Source
M_BH	$1.26 \times 10^6 \text{ kg}$ ($6.5 \times 10^6 M_\odot$)	EHT 2019 (first image)
r_Schwarzschild	$1.92 \times 10^{13} \text{ m}$	$2GM/c^2$
r_horizon (UQFF)	$1.95 \times 10^{13} \text{ m}$	$r_S \times (1 + 0.015)$
Distance	$16.8 \text{ Mpc} = 5.18 \times 10^{23} \text{ m}$	Virgo Cluster
Spin (a/M)	0.90 ± 0.05	EHT 2024
Jet power P_jet	$\sim 10^{44} \text{ erg/s}$	VLA/VLBI
T_H (GR)	$1.35 \times 10^{17} \text{ K}$	$?c^3/(8\pi GMk_B)$
T_UQFF	$1.34 \times 10^{17} \text{ K}$	$T_H \times 0.99$

2. 8-Term MUGE at $r_{\text{horizon}} = 1.95 \times 10^{13} \text{ m}$

Term	Value (m/s 2)	Notes
base_gravity	2207	Newton dominant

Term	Value (m/s ²)	Notes
sum_Ug	3.75	Ug4 ? M ² /r ⁶ ? M87 larger r offsets large M ²
U_i	0.14	
cosmological	-9.1×10^{21}	? negligible at horizon
quantum	$+2.0 \times 10^{41}$	Planck-scale
fluid	$+6.2 \times 10^{13}$	Jet plasma viscosity
dark_matter	+0.044	Virgo cluster DM halo
coherence	peaked at horizon	Gaussian, » far_field
g_total	2211	100%

Newtonian g_Newton = 2207 m/s². MUGE total = 2211 m/s² ? UQFF excess = +0.18%.

3. Jet Power: Ug3 UQFF Mechanism

The M87 jet (1.4 kpc visible, Lorentz factor $\Gamma \approx 6$) is mediated by Ug3 string rotation in the UQFF:

$$P_{\text{jet}}^{\text{UQFF}} = U_{g3}(r_{\text{ISCO}}) \cdot \dot{M}_{\text{acc}} c^2 \cdot [\text{SCm}]$$

With $\eta = \eta_{\text{acc}}/\eta_{\text{Edd}} \approx 10^{-3}$ (low state) and $[\text{SCm}] = 0.99$:

$$P_{\text{jet}}^{\text{UQFF}} = 0.99 \times 10^{-3} L_{\text{Edd}} = 3.6 \times 10^{44} \text{ erg/s}$$

This suggests UQFF jet efficiency $\eta_{\text{jet}} = 0.99 \times 0.001 = 0.099\%$, consistent with M87 observational estimates of $\sim 0.1\%$ radiative efficiency in FR I jets.

4. Shadow Diameter Cross-Check

EHT observed ring diameter: $\theta_{\text{ring}} = 42 \pm 3 \mu\text{as}$? physical $r_{\text{ring}} = 5.0 \text{ GM}/c^2$ (photon ring).

UQFF prediction: The UQFF slightly shifts the photon capture cross-section via $[\text{SCm}]$:

$$r_{\text{shadow}}^{\text{UQFF}} = r_{\text{shadow}}^{\text{GR}} \cdot \sqrt{1 + \frac{1 - [\text{SCm}]}{2}} = r_{\text{GR}} \times \sqrt{1.005} \approx 1.0025 r_{\text{GR}}$$

$\Delta\theta = +0.25\% \approx 0.1 \mu\text{as}$ shift (undetectable by current EHT at $\pm 3 \mu\text{as}$ precision).

5. Coherence vs Distance

At M87* with its much larger r_{horizon} (vs Sgr A*):

Location	r/r_{horizon}	g_{coh}	Ratio
At horizon (1.95×10^{13} m)	1.0	$g_{\text{coh},0}$	1.000
1 kpc (3.1×10^{17} m)	1.6×10^6	~ 0	$\sim 10^{-6}$

From validator: `assert coh_at_horizon > coh_far * 1e6` **PASS** for M87* system.

6. Hawking Temperature and UQFF Ratio

$$T_H^{\text{M87*}} = \frac{\hbar c^3}{8\pi G M k_B} = \frac{1.055 \times 10^{-34} \times (3 \times 10^8)^3}{8\pi \times 6.674 \times 10^{-11} \times 1.26 \times 10^{40} \times 1.38 \times 10^{-23}}$$

$$= 1.35 \times 10^{-17} \text{ K}$$

$$T_{\text{UQFF}}^{\text{M87*}} = 0.9999 \times T_H = 1.34 \times 10^{-17} \text{ K}$$

$\delta = 0.01\%$ reduction from GR. IceCube and FRB backgrounds: consistent.

Summary

Validation	Result
All 8 MUGE terms finite	δ PASS
$g_{\text{total}} = \text{Newton} + 0.18\%$	δ PASS
No NaN/Inf for M87*	δ PASS
Coherence peak at horizon	δ PASS
Jet power UQFF estimate	3.6×10^{44} erg/s (consistent)
Shadow diameter deviation	0.25% ($\ll$ EHT precision)
T_{UQFF}	1.34×10^{-17} K

Source: `validate_uqff_muge.py` | `from_system('M87')` | EHT 2019-2024 | $[SCM]=0.99$
 | all 8 terms PASS

UQFF computed: Canonical UQFF buoyancy parameter $U_{bi} = \frac{1}{2} \times [SSq] \times GM/r^2 = 5.0e-4 \times 0.57 \times 6.67e-11 \times M/r^2$; for solar parameters: $U_{bi,Sun} = 5.7e-4 \times 6.67e-11 \times 1.99e30 / (6.96e8)^2 = 1.47e+2 \text{ m/s}^2$.

Appendix: UQFF Production Framework Reference (v4.75+)

Added by upgrade_early_whitepapers.py (v4.75). This appendix cross-references the production physics constants and master equations to enable reproducibility against the current codebase state.

A.1 Calibration Constants

Symbol	Value	Description
$\hat{\Gamma}^0$	$5.0 \times 10^{-4} \text{ day}^{-1}$	UQFF exponential decay rate
[SSq]	0.57	Universal Quantized Factor
$\hat{\Gamma}^2_i$	0.60 ± 0.61	Buoyancy coupling coefficient
k_{Ug1}	1.5	Ug1 DPM-dipole coupling
k_{Ug2}	1.2	Ug2 outer-bubble charge coupling
k_{Ug3}	1.8	Ug3 string-rotation coupling
k_{Ug4}	2.0	Ug4 vacuum-concentration coupling
$\hat{I} \cdot$	10 Å^2	Inertia tensor scale
$E_{\text{react}}(0)$	$10 \text{ Å}^2 \text{ J}$	Reference reactive energy

A.2 F_U Master Equation (Complete $\hat{\Gamma}^2_i$ 4 terms)

$$F_U = U_{g1} + U_{g2} + U_{g3} + U_{g4} + U_{bi} + U_m - \sum_{i=1}^4 \text{igl}[\lambda_i \cdot U_i(r, t) \cdot E_{\text{react}} \text{igr}]$$

Term	Description	Implementation
Ug1	DPM magnetic dipole	compute_Ug1_SOURCE4 / compute_Ug1()
Ug2	Outer-field bubble (charge-reactivity)	compute_Ug2_SOURCE4 / compute_Ug2()
Ug3	Magnetic string rotation	compute_Ug3_SOURCE4 / compute_Ug3()
Ug4	Vacuum concentration (star-BH)	compute_Ug4_SOURCE4 / compute_Ug4()
Ubi	Buoyancy force	compute_Ubi_SOURCE4 / compute_Ubi()
Um	Universal Magnetism (Heaviside-amplified)	compute_Um_SOURCE4 / compute_Um()
$\hat{\Gamma}^2_i \hat{\Gamma}^0$	4th-order dissipation term (PAPER_420)	compute_FU_SOURCE4 / full pipeline

4th dissipation term parameters (PAPER_420):

$\hat{I}_{\text{c}}=10\hat{\mu}\gg\hat{A}^1\hat{\mu}^\circ$, $\hat{I}_{\text{q}}=10\hat{\mu}\gg\hat{A}^1\hat{A}^2$, $\hat{I}_{\text{a}}=10\hat{\mu}\gg\hat{A}^1\hat{A}^1$, $\hat{I}_{\text{b}}=10\hat{\mu}\gg\hat{A}^1\hat{A}^3$ (free parameters, not yet empirically calibrated)

A.3 Um Heaviside Phase-Transition Amplifier (PAPER_421)

$$U_m^{\text{full}} = U_m^{\text{base}} \text{imesigl}(1 + 10^{13} \Theta(ho_{SCm} - ho_c)igr) \text{imesigl}(1 + A_q \cos(\Delta\omega t)igr)$$

Symbol	Value	Description
$\hat{I}_{\text{c}}$	$10\hat{A}^1\hat{\mu}\text{ kg/m}\hat{A}^3$	SCm critical superconducting density
A_q	0.1	Quasi-periodic beating amplitude (10%)
$\hat{I}_{\text{b}}$	$2\hat{I}_{\text{b}}/(434\hat{A}\cdot365.25)\text{ rad/day}$	434-year Gleisberg supercycle

A.4 UQFF Four Operational Modes

Mode	Dominant Term	Primary Use Case
Compressed	Ug_{sum} + Newtonian base	Isolated stellar/BH systems
Resonant	5 resonance frequencies (aDPM, aTHz, $\hat{\mu}^\circ$)	Multi-scale field interactions
Buoyant	$\hat{I}_{\text{c}}^2 \hat{I}_{\text{b}} \hat{A} \hat{I}_{\text{a}}$	Expanding nebulae, stellar winds
Superconductive	$\hat{I}_{\text{c}} \hat{A} (1 + 10\hat{A}^1\hat{A}^3\hat{A}\cdot f_{\text{H}})$	Magnetars, SCm critical-density regime

Implementation status: all 4 modes operational in MAIN_1_CoAnQi.cpp, CondensedPhysics.py, and CondensedPhysics2.py.